

THE LIBERATOR'S DILEMMA

Game Theory, Power & the Independence of Latin America

A Plain-Language Guide to the Academic Paper

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"No one could understand how such a frail man had the energy not only to lead his soldiers through three countries but also to reason and write as much as he did."

— García Márquez, The General in His Labyrinth

1. What Is This Paper About?

Imagine you are playing chess — but instead of pieces on a board, the pieces are armies, colonial governors, creole landowners, and hundreds of thousands of people across an entire continent. The moves are battles, political alliances, promises, and betrayals. The stakes are the freedom of six nations.

That, in essence, is what this paper does. It takes one of history's most dramatic stories — Simón Bolívar's liberation of South America between 1810 and 1826 — and analyses it using game theory: the mathematical science of strategic decision-making.

The paper draws on two main inspirations. The first is Gabriel García Márquez's Nobel-winning novel *El general en su laberinto*, following Bolívar's final journey down the Magdalena River in 1830, sick and watching his republic crumble. The second is Acemoglu & Robinson (2005): a landmark book asking why some countries become democracies and others dictatorships, answered through strategic power games between elites and citizens.

2. What Is Game Theory?

Game theory is the study of how rational people make decisions when the outcome for each person depends on what others decide. Three concepts are especially important here:

- **Nash Equilibrium:** A stable situation where no player can improve their outcome by changing strategy alone. A resting point in the game.
- **Credible Commitment:** A promise that is believable because breaking it would be too costly. A king who promises reforms but can reverse them once the crisis passes makes an incredible promise.
- **Subgame-Perfect Equilibrium:** The best possible decision at every stage of a sequential game — not just at the start, but at each branch of the decision tree.

3. Bolívar's Campaigns: Six Nations, Sixteen Years

Bolívar did not liberate one country. He liberated six: Venezuela, Colombia, Ecuador, Peru, Bolivia, and Panama. He marched armies across the Andes in winter, crossed fever-ridden jungles, and fought for sixteen years. The paper's TikZ map traces these routes:

Campaign	Year	Key Battle	Result
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Admirable Campaign	1813	Caracas	First victories in Venezuela
Andes Crossing	1819	Boyacá	Liberation of Colombia
Carabobo Campaign	1821	Carabobo	Final liberation of Venezuela
Ecuador Campaign	1822	Pichincha	Liberation of Ecuador
Peru Campaign	1824	Junín & Ayacucho	Liberation of Peru
Sucre / Upper Peru	1825	Tumusla	Creation of Bolivia

4. The Nine Strategic Games

Game 1: The Caracas Junta (1810) — A Crisis Creates an Opportunity

When Napoleon occupied Spain, the Spanish Crown lost authority. Creole elites in Caracas faced a choice: obey a phantom king, or seize the moment. The crisis made repression too costly. Forming a junta was the rational and historical outcome.

GAME: Summary

Type: Sequential Game. Outcome: Junta formation. Key insight: External shocks can suddenly change which strategy is rational.

Game 2: The First Republic — A Coordination Failure (1811–1812)

The First Venezuelan Republic collapsed because patriots failed to coordinate. Different regions preferred dominance over unity. The paper models this as a Stag Hunt: cooperation pays off, but only if everyone cooperates. Everyone defected. The republic fell.

GAME: Summary

Type: Stag Hunt. Outcome: Defection — republic collapses. Key insight: Good intentions are not enough; coordination requires trust and sometimes an enforcer.

Game 3: Recruiting the Llaneros — Costly Signals (1817–1818)

The llaneros were fierce plainsmen, initially royalist. Bolívar needed them. Since words are cheap, he used costly signals: land grants, military promotions, the guerra a muerte decree. Expensive commitments that a half-hearted leader could not afford to make.

GAME: Summary

Type: Signalling Game. Outcome: Separating equilibrium — Bolívar credibly signals commitment. Key insight: Expensive actions communicate sincere intentions when words alone cannot.

Game 4: The Angostura Congress (1819)

Convening a constitutional congress under military threat, Bolívar found that external royalist pressure made rejection by congress too costly. The threat tilted the equilibrium toward acceptance of his constitutional proposals.

GAME: Summary

Type: Extensive Form Game. Outcome: Constitution accepted. Key insight: External threats can make internal compromise rational even between rivals.

Game 5: Boyacá — The Point of No Return (1819)

Crossing the Andes in winter with 2,000 soldiers was a commitment device. Once crossed, retreat was impossible. The royalists could not credibly negotiate because Bolívar had no choice but to fight. When retreat is impossible, the enemy's best response is often surrender.

GAME: Summary

Type: Irreversible Commitment Game. Outcome: Royalist retreat, liberation of Colombia. Key insight: Destroying your own option to retreat can make you stronger.

Game 6: Guayaquil — The Secret Meeting (1822)

Bolívar met San Martín in July 1822. No one knows what was said. San Martín left, resigned, and went into exile. The paper models this as a Nash Bargaining Solution: given Bolívar's greater resources and political capital, the fair split required San Martín to cede command. He did. It was rational.

GAME: Summary

Type: Nash Bargaining Game. Outcome: San Martín cedes command. Key insight: Power asymmetry produces predictable outcomes, even in secret meetings.

Game 7: Peru — When Elites Switch Sides (1823–1824)

Peru's creole elite feared social revolution more than Spanish rule. As Spanish authority collapsed, its loyalty offer became worthless. At a critical tipping point — shown in a graph in the paper — siding with Bolívar became more profitable than loyalty to the Crown.

GAME: Summary

Type: Redistribution / Threshold Game. Outcome: Elite defection, royalists defeated. Key insight: Elites switch sides when the payoff calculation changes, not for ideology.

Game 8: Ayacucho — End of the Game (December 9, 1824)

By Ayacucho, the game was structurally over. Spain had lost its credibility, its elite allies, and its commitment technology. The royalist army's dominant strategy was negotiated surrender. The paper calls this the terminal node of the entire liberation game.

GAME: Summary

Type: Terminal Node / SPNE. Outcome: Spanish surrender, South American independence. Key insight: Some games end not with a bang but with a calculation.

Game 9: Gran Colombia — Why Bolívar's Republic Fell Apart (1826)

Gran Colombia — Bolívar's unified republic spanning Venezuela, Colombia, and Ecuador — lasted less than a decade. Each member state calculated that going it alone was better than federation. The 'core' of the cooperative game was empty; dissolution was inevitable.

GAME: Summary

Type: Cooperative Game / Empty Core. Outcome: Dissolution into three republics. Key insight: Political unions collapse when individual parts gain more outside than inside.

5. García Márquez: The Novelist as Political Economist

The paper weaves eight novels by Gabriel García Márquez into the analysis, arguing that great literature often captures the same truths as formal models — expressed in human rather than mathematical language.

Novel	Year	Political Economy Theme
One Hundred Years of Solitude	1967	Extractive institutions repeating across generations
The Autumn of the Patriarch	1975	The extractive equilibrium: a dictator no one can coordinate to overthrow
Love in the Time of Cholera	1985	Social immobility as a Nash Equilibrium
The General in His Labyrinth	1989	Coalition collapse; Gran Colombia's empty core
No One Writes to the Colonel	1961	The state's broken promise as a one-sided game
In Evil Hour	1962	Local power and the economics of repression
Of Love and Other Demons	1994	Church vs. civil society as competing institutions
Memories of My Melancholy Whores	2004	Time preference and the cost of deferring life's choices

6. The Big Takeaways

“Bolívar solved the military game of liberation — and lost the political game of institution-building.”

- Colonial regimes collapse when their commitment technology is destroyed. Napoleon's occupation was that shock for Latin America.
- Costly signals — not words — build coalitions. Bolívar won the llaneros through concrete, expensive commitments.
- Guayaquil was a rational bargain, not a mystery. Power asymmetry produces predictable outcomes even in secret meetings.
- Post-independence failure was built-in. When the elites who made independence possible control the new states, extractive institutions persist.
- Gran Colombia's collapse was mathematically inevitable. A coalition with an empty core always breaks apart.

García Márquez called this condition *soledad* — solitude. The isolation of societies unable to coordinate on the institutions that would make them free. It is, the paper argues, not a metaphor. It is a Nash Equilibrium.

7. Want to Go Deeper?

Book	Author	Why Read It
The General in His Labyrinth	García Márquez	Literary heart of this paper — Bolívar's final days
Economic Origins of Dictatorship...	Acemoglu & Robinson	The formal framework; accessible from ch.2
Why Nations Fail	Acemoglu & Robinson	Popular version of the same ideas
Simón Bolívar: A Life	John Lynch	The definitive biography
One Hundred Years of Solitude	García Márquez	Greatest Latin American novel
A Course in Game Theory (free online)	Osborne & Rubinstein	Best introduction to formal game theory

— Miguel Rocha de Sousa | Universidade de Évora, Department of Economics, Portugal